
Exercise (Recitation Sep 24). Consider the function

$$f(x) = \int_0^1 \frac{1 - e^{-xt^2}}{t} dt.$$

1. Find the set D of x for which $f(x)$ is well-defined and real-valued.
2. Prove that $f \in C^1(D)$.
3. Compute $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$.
4. Prove that the function

$$F(x, y) = \int_0^1 \frac{1 - e^{-xyt^2}}{t} dt$$

is C^1 in an open set of $\mathbb{R}^2$.

5. Compute

$$\lim_{(x,y) \rightarrow (0,0)} \frac{F(x, y)}{\sqrt{x^2 + y^2}}.$$

Proof. 1. $D = \mathbb{R}$. To show this, fix x and let

$$g(t) = \frac{1 - e^{-xt^2}}{t}.$$

First show that $g(t) \rightarrow 0$ as $t \rightarrow 0$. With Taylor's theorem, we have

$$e^{-xt^2} = 1 - xt^2 + o(t^3).$$

Thus $1 - e^{-xt^2} = xt^2 + o(t^3)$. Note that the little o here is considering the limit as $t \rightarrow 0$. Therefore, $g(t) \rightarrow 0$ as $t \rightarrow 0$.

Since g is continuous on $(0, 1]$, it is measurable. Now if $g(t) \rightarrow 0$ as $t \rightarrow 0$, we can show $f(x) < \infty$ for any $x \in \mathbb{R}$. Indeed, by the limit above, there exists $0 < \delta < 1$ such that $g(t) < 1$ on $(0, \delta]$. Since g continuous $g(t)$ is bounded on compact interval $[\delta, 1]$. Hence, $g(t)$ is bounded on $(0, 1]$, and thus $f(x) < \infty$ for all $x \in \mathbb{R}$.

2. First show that the derivative of f is

$$T(x) = \int_0^1 te^{-xt^2} dt.$$

This is well defined, as for each fixed x , we have

$$te^{-xt^2} \leq \max\{e^{-x}, 1\}$$

for $t \in [0, 1]$. Therefore, $T(x) < \infty$.

Fix $x \in \mathbb{R}$. We want to show

$$\lim_{u \rightarrow 0} \frac{|f(x+u) - f(x) - T(x)u|}{|u|} = 0.$$

Let $\{u_n\}_{n=1}^{\infty}$ be any sequence with $u_n \rightarrow 0$ and $|u_n| \leq 1$. We have

$$\frac{|f(x+u_n) - f(x) - T(x)u_n|}{|u_n|} = \frac{1}{|u_n|} \int_0^1 \left| \frac{e^{-xt^2} - e^{-(x+u_n)t^2}}{t} - te^{-xt^2} u_n \right| dt.$$

By mean value theorem we have

$$\left| \frac{e^{-xt^2} - e^{-(x+u_n)t^2}}{t} - te^{-xt^2} u_n \right| \leq |u_n| \sup_{\xi \in [0, u_n]} \left| te^{-(x+\xi)t^2} - te^{-xt^2} \right|.$$

Then,

$$\frac{|f(x+u_n) - f(x) - T(x)u_n|}{|u_n|} \leq \int_0^1 \sup_{\xi \in [0, u_n]} \left| te^{-(x+\xi)t^2} - te^{-xt^2} \right| dt.$$

Towards using DCT, note that

$$\left| te^{-(x+\xi)t^2} - te^{-xt^2} \right| \leq 2t \max\{e^{-x+1}, 1\} \leq 2 \max\{e^{-x+1}, 1\}$$

for $t \in [0, 1]$. Note that the RHS is a constant w.r.t. t and thus integrable on $[0, 1]$. By DCT, this implies

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{|f(x+u_n) - f(x) - T(x)u_n|}{|u_n|} &\leq \lim_{n \rightarrow \infty} \int_0^1 \sup_{\xi \in [0, u_n]} \left| te^{-(x+\xi)t^2} - te^{-xt^2} \right| dt \\ &= \int_0^1 \lim_{n \rightarrow \infty} \sup_{\xi \in [0, u_n]} \left| te^{-(x+\xi)t^2} - te^{-xt^2} \right| dt \\ &= 0, \end{aligned}$$

since $u_n \rightarrow 0$ as $n \rightarrow \infty$, $\xi \in [0, u_n]$, and $\xi \mapsto te^{-(x+\xi)t^2}$ is continuous.

Since $\{u_n\}_{n=1}^{\infty}$ is arbitrary sequence converges to 0, this shows the desired limit and $Df = T$.

To show $f \in C^1(\mathbb{R})$, it remains to show T is continuous, i.e. $T(x+u) - T(x) \rightarrow 0$ as $u \rightarrow 0$.

For this again let $\{u_n\}_{n=1}^{\infty}$ be any sequence with $u_n \rightarrow 0$ and $|u_n| \leq 1$. We have

$$T(x+u_n) - T(x) \leq \int_0^1 \left| te^{-(x+u_n)t^2} - te^{-xt^2} \right| dt.$$

Again note

$$\left| te^{-(x+u_n)t^2} - te^{-xt^2} \right| \leq 2t \max\{e^{-x+1}, 1\} \leq 2 \max\{e^{-x+1}, 1\}$$

for $t \in [0, 1]$, which is integrable on $[0, 1]$. Therefore, by DCT,

$$\begin{aligned} \lim_{n \rightarrow \infty} T(x + u_n) - T(x) &\leq \lim_{n \rightarrow \infty} \int_0^1 \left| te^{-(x+u_n)t^2} - te^{-xt^2} \right| dt \\ &= \int_0^1 \lim_{n \rightarrow \infty} \left| te^{-(x+u_n)t^2} - te^{-xt^2} \right| dt \\ &= 0, \end{aligned}$$

since $u_n \rightarrow 0$ as $n \rightarrow \infty$, and $u \mapsto te^{-(x+u)t^2}$ is continuous.

Since $\{u_n\}_{n=1}^\infty$ is arbitrary sequence converges to 0, this shows the desired limit and completes the proof. $\square$

Remark. Some useful forms of mean value theorem: Let X and Y be normed vector space and $U \subset X$ open. Suppose that $x, y \in U$ with $x \neq y$ and that

$$L(x, y) = \{\alpha y + (1 - \alpha)x : \alpha \in [0, 1]\} \subset U.$$

Suppose that $f : U \rightarrow Y$ is continuous on $L(x, y)$ and differentiable on $L(x, y) \setminus E$, where $E \subset L(x, y)$ is countable. Then the following holds:

1. If

$$\sup_{z \in L(x, y) \setminus E} \|Df(z)\| = M < \infty,$$

then

$$\|f(y) - f(x)\| \leq M \|x - y\|.$$

2. If $T \in \mathcal{L}(X; Y)$ and

$$\sup_{z \in L(x, y) \setminus E} \|Df(z) - T\| = K < \infty,$$

then

$$\|f(y) - f(x) - T(y - x)\| \leq K \|x - y\|.$$

Exercise (Midterm 1). Let

$$f_n(x) = nx^2(1 - x)^n \cos(2\pi x), \quad x \in [0, 1].$$

1. Does the series of functions $\sum_{n=1}^\infty f_n$ converges pointwise on $[0, 1]$.
2. Does the equality

$$\sum_{n=1}^\infty \int_0^1 f_n dx = \int_0^1 \sum_{n=1}^\infty f_n dx$$

hold?

3. Does the series of function $\sum_{n=1}^{\infty} f_n$ converges uniformly on $[0, 1]$?

Proof. 1. Yes. The series clearly converges at 0 and 1. For $0 < x < 1$, we have

$$|f_n(x)| \leq nx^2(1-x)^n.$$

Notice that $0 < 1-x < 1$, so the series converges by root test.

2. Yes. Notice that

$$\int_0^1 |f_n| \leq \int_0^1 nx^2(1-x)^n = \frac{2n}{(n+1)(n+2)(n+3)}.$$

Therefore,

$$\int_0^1 \left| \sum_{n=1}^{\infty} f_n \right| \leq \int_0^1 \sum_{n=1}^{\infty} |f_n| = \sum_{n=1}^{\infty} \int_0^1 |f_n| < \infty$$

The desired equality then follows from DCT.

3. No. For $N, M \in \mathbb{N}$, consider

$$\sum_{n=N}^M f_n(x) = \sum_{n=N}^M nx^2(1-x)^n \cos(2\pi x).$$

Note that $x \mapsto x^2(1-x)^n$ attains its maximum at $\frac{2}{2+n}$, so consider

$$\begin{aligned} \left\| \sum_{n=N}^M f_n \right\| &\geq \sum_{n=N}^M f_n \left(\frac{2}{2+n} \right) \\ &= \sum_{n=N}^M \frac{4n}{(n+2)^2} \left(\frac{n}{n+2} \right)^n \cos \left(\frac{4\pi}{n+2} \right) \\ &\geq \sum_{n=N}^M \frac{n}{(n+2)^2} \frac{1}{e^2} \end{aligned}$$

for N large enough. However note that $\sum_{n=1}^{\infty} \frac{n}{(n+2)^2}$ diverges, so the partial sums is not Cauchy. It follows that $\sum_{n=1}^{\infty} f_n$ is not uniformly Cauchy either, so the series of function does not converge uniformly on $[0, 1]$.

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